EXP 2: To implement all process management commands

1) bg : It is used to resume a suspended (stopped) process and run it in the background

```
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ bg 1
bash: bg: job 1 already in background
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ bg 2
bash: bg: job 2 already in background
```

2) Pidof : It is a simple utility used to find the **Process ID (PID)** of a running program by its name.
3) Ps : It provides a detailed snapshot of the currently active processes on a system.

```
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ pidof firefox
8755 8564
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ ps 8755
    PID TTY      STAT   TIME COMMAND
   8755 pts/0    S      0:00 /snap/firefox/7836/usr/lib/firefox/firefox -contentproc -ipcHandle 0 -signalPipe 1 -initialChannelId {76135b51-7622
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$
```

4) &function : It runs a command in the background, allowing you to continue using the terminal while the job runs.

```
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ firefox &
[1] 8564
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ Gtk-Message: 14:25:40.384: Not loading module "atk-bridge": The functionality is provide
[8564, Main Thread] WARNING: GTK+ module /snap/firefox/7836/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so canno
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.: 'glib warning', file /build/firefox/pa

(firefox_firefox:8564): Gtk-WARNING **: 14:25:40.448: GTK+ module /snap/firefox/7836/gnome-platform/usr/lib/gtk-2.0/modules/libcan
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.
Gtk-Message: 14:25:40.448: Failed to load module "canberra-gtk-module"
[8564, Main Thread] WARNING: GTK+ module /snap/firefox/7836/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so canno
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.: 'glib warning', file /build/firefox/pa

(firefox_firefox:8564): Gtk-WARNING **: 14:25:40.449: GTK+ module /snap/firefox/7836/gnome-platform/usr/lib/gtk-2.0/modules/libcan
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.
Gtk-Message: 14:25:40.449: Failed to load module "canberra-gtk-module"
^C
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ pidof firefox
8755 8564
```

5) Nice-renice : used to manage the scheduling priority of processes

```
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ nice -n 20 libreoffice
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ renice 17 -p 6238
6238 (process ID) old priority 0, new priority 17
```

6) fg : used to move a background or stopped job into the foreground of the current terminal session

```
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ libreoffice
Warning: failed to launch javaldx - java may not function correctly
^C
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ libreoffice &
[1] 7027
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ Warning: failed to launch javaldx - java may not function correctly
^C
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ fg
libreoffice
```

7) Top : used for real-time system monitoring. It provides a dynamic, continuously updated view of running processes, CPU usage, and memory consumption

```
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ top

top - 14:00:45 up 22 min,  2 users,  load average: 0.15, 0.21, 0.18
Tasks: 324 total,   1 running, 323 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.1 us,  0.1 sy,  0.0 ni, 99.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  15616.9 total,  11700.4 free,   2220.0 used,   2386.3 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.  13396.9 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   6471 lab-03-+  20   0 2677712 177572 101628 S   0.7   1.1   0:02.14 Privileged Cont
      7 root      20   0       0      0      0 I   0.4   0.0   0:00.05 kworker/0:0-events
   2309 lab-03-+  20   0 5949428 285572 118176 S   0.4   1.8   0:18.96 gnome-shell
   3783 lab-03-+  20   0  775388  57268  43888 S   0.4   0.4   0:03.02 gnome-terminal-
      1 root      20   0  171112  14448   8944 S   0.0   0.1   0:02.18 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_flushwq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
     11 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     13 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     15 root      20   0       0      0      0 S   0.0   0.0   0:00.01 ksoftirqd/0
     16 root      20   0       0      0      0 I   0.0   0.0   0:00.35 rcu_preempt
     17 root      rt   0       0      0      0 S   0.0   0.0   0:00.00 migration/0
     18 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     21 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
     22 root      rt   0       0      0      0 S   0.0   0.0   0:00.15 migration/1
     23 root      20   0       0      0      0 S   0.0   0.0   0:00.01 ksoftirqd/1
     25 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
     26 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/2
     27 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/2
     28 root      rt   0       0      0      0 S   0.0   0.0   0:00.15 migration/2
```